

Owen Peckham MEng

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First-Class Mechanical & Electrical Engineering graduate and **current PhD Researcher** specialising in advanced structural optimisation, additive manufacturing, and high-fidelity systems prototyping. Proven track record of delivering quality research and design outputs. Experienced in cross-disciplinary project management and full-lifecycle engineering design from concept to physical fabrication. Seeking to leverage my expertise in cutting-edge computational mechanics and robust system design to drive innovation in the UK space sector.

Key Skills

(A): Advanced, (W): Working

Category	Skill(s)	Experience Example
CAD	Autodesk Fusion (A), Solidworks (W)	Design of rigs for holding samples during destructive testing
Programming & Computing	Python (A), GPU Numba (A), C++ (W), LaTeX (W), Slurm (W)	Development of novel GPU-based MINLP optimisation methods
Engineering Tools	Abaqus/CAE (A), MATLAB & Simulink (A), Anylogic (W)	Simulation of topology optimised test artefact performance
Practical Manufacturing	Rapid Prototyping (A), Turning & Milling (W), Composite layup (W)	Machining custom electric car wheel hubs

Leadership Experience

Co-Founder, Co-President and Technical Manager, Bristol Racing Society *2021–2023*

- **Co-Founded Bristol Racing**, a society that builds small electric vehicles for endurance races. Grew the society to **94 paying members**.
- **Re-elected** as co-president in 2022. Oversaw a **45 second reduction in lap time** from the first to second year.
- Attained **top 5/25** against teams from for example Jaguar Land Rover and Renishaw.
- **Technical Manager** (2022/23) - I managed the progress of all the sub-teams, advising on design and manufacturing decisions and directing the overall schedule.
- Chassis and mechanical **sub-team lead** (2021/22) - I led the design and manufacture for the sub-team, managing members and ensuring that progress was maintained.
- **Negotiated the university's first** student self-supervised after-hours workshop access scheme.
- Awarded **£8,800** from funding grants over two years.
- Built partnerships with industry, including the **National Composites Centre**, who provided workshop space, funding, and resources.

Relevant Employment History

PhD Researcher, DMF Lab, The University of Bristol *2023–date*

- Thesis title: “Improving Generative Design: An Investigation Into Strategies For Lightweighting Using Hollow Structures”.
- Performed the first demonstration of hybrid level set and phase field topology optimisation in 3D.
- Developed new approaches to structural lightweighting that **increase the load capacity of topology optimised outcomes by >20%**.
- Worked with others in the DMF on projects related to my main research that address additive manufacturing, topology optimisation and generative design, and AI.
- **Scholarship recipient** for the Arabic graduate language training programme awarded a 2:1.
- Undertook a 3-month research trip to the Integrated Design Innovation Group at Carnegie Mellon University, USA. From the collaboration we have **developed a novel GPU-based MINLP optimiser capable of solving previously uncracked benchmark problems**. Currently finalising the manuscript of an article targeting the Journal of Engineering Design as first author.

Graduate Teacher, The University of Bristol

2023–date

- **Developed and deliver a yearly masters-level guest lecture** on the technical specifics of level set topology optimisation.
- Taught, marked assessments and ran practical and computer labs for 3 undergraduate units in mechanical engineering and electrical machine design. Acted as a **group manager for 48 students per year**.

Research Engineer, NDT Lab, The University of Bristol

Summer 2024–Spring 2025

- Optimised the manufacturing procedures of low-cost additively manufactured microfluidics.
- Developed an Autodesk Fusion add-in tool that automates the design of the microfluidic scaffolds.
- **Invited to deliver a 2-day technical workshop** on microfluidic manufacturing methodologies at the Chulabhorn Research Institute (Bangkok, Thailand).

Research Internship, DMF Lab, The University of Bristol

Summer 2022

- Investigated the impact of information sharing within distributed decentralised agent based additive manufacturing networks.
- Used AnyLogic to simulate the network and created a classification system that enabled robust parameterisation of the necessary communication.

Internship, Cambustion Ltd.

Summer 2021

- Designed and built an Automatic Number Plate Recognition device that could be triggered by one of Cambustion's roadside gas analysers. I implemented the ANPR functionality using a Raspberry Pi 4 with a camera, using APIs from Google (text recognition) and the UK Drivers & Vehicles Standards Agency.
- Designed and built a test rig that enabled engineers to test the leakage rate of pressurised fuel injectors. Used another Pi 4 for the controller and developed a cloud-linked spreadsheet as a user interface.

Education

The University of Bristol, Undergraduate

2019–2023

- Mechanical and Electrical Engineering, MEng - **awarded first class honours**.
- **Awarded first class honours and joint highest mark in cohort** for my fourth-year group industrial project: "The Canute Braille Reader: Motor Analysis, Control Methods, and Alternative Solutions". Fundamentally redesigned the partner company's multi-line braille e-reader, building a novel approach that enabled **infinite reading without page refresh delays**.
- **Awarded first class honours** for my third-year individual research project: "An investigation into the use of least-squares system-identification and analytical methods to tune a Proportional-Integral controller for a Buck converter with a resistive load". Devised a method of using least-squares regression to determine the component sizes in a Buck converter. Using this knowledge my automated controller tuner successfully managed the steady-state error and rise times of the converter's step response to **within 5% of the desired values**.

Research Output & Publication Experience

- Published **2 journal** and **3 conference papers** as first author and 2 conference papers as a contributing author. Awarded editor's choice for one of the journal papers.
- **Presented at 3 international engineering conferences**.
- Reviewed 2 papers for Springer Nature's 'Scientific Reports' — **the third-most cited journal in the world**.

References available upon request.